

MAKE UP EXAMINATIONS – AUGUST 2024

Program	: B.E. – Information Science and Engineering	Semester	: VI
Course Name	: Machine Learning	Max. Marks	: 100
Course Code	: IS62	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Discuss the types of Machine Learning. CO1 (06)
 - In a Spam Mail Filter Classification Model to filter out spam 5000 email samples were examined. The number of TP = 800, FP = 500, TN = 3500, FN = 200. Do the following for this Spam Mail Filter Classification Model:
 - Draw the Confusion Matrix
 - Calculate the Accuracy value
 - Assess the Sensitivity Value
 - Estimate the Specificity Value
 - Measure the Error Rate
 - Determine the Precision Value
 - Compute the Recall Value
 - Evaluate the F-Measure.
 - Discuss how Bias-Variance Tradeoff concept is associated with underfitting and overfitting in Machine Learning. CO1 (06)
- Illustrate the process of machine learning with a neat diagram. CO1 (07)
 - Consider the given confusion matrix and compute the following measures: CO1 (06)
 - Accuracy
 - True positive and True Negative,
 - False positive and False negatives,
 - Sensitivity value.
 - Specificity value.
 - Precision and Recall.

		Predicted		
		Cat	Lion	Tiger
Actual	Cat	10	1	2
	Lion	1	8	2
	Tiger	3	1	6

- What do you mean by Imbalanced Datasets? With relevant examples explain why Accuracy fails as a performance metrics for these datasets? CO1 (07)

UNIT - II

- Find dissimilarity matrix for the given dataset consisting of mixed attribute. CO2 (12)

ID	Test-1 Nominal	Test-2 Ordinal	Test-3 Numeric
1	Code A	Excellent	45
2	Code B	Fair	22
3	Code C	Good	64
4	Code A	Excellent	28

- Give any two reasons with examples why we need to do "dimensionality reduction". Explain LDA with an example. CO2 (08)

4. a) Explain the following with an example. CO2 (08)
- Nominal attributes
 - Ordinal attributes
 - Binary attributes
 - Numeric attribute.
- b) Find dissimilarity matrix for the given dataset consisting of numeric CO2 (04)
attribute. Consider Euclidean distance to calculate distance matrix.

	A1	A2
P1	0	2
P2	2	0
P3	3	1
P4	5	1

- c) Consider the two dimensional patterns (2, 1), (3, 5), (4, 3), (5, 6), (6, 7), CO2 (08)
(7, 8). Compute the principal component using PCA Algorithm.

UNIT - III

5. a) Below is a dataset of the outcome of testing with a group of people, if a CO3 (07)
type of tissue paper is good or not. Apply **K-NN to classify** if a new
tissue with Acid Durability 3 and Strength 7 is Good or Bad.

Name	Acid Durability	Strength	Class
Type-1	7	7	Bad
Type-2	7	4	Bad
Type-3	3	4	Good
Type-4	1	4	Good

- b) Apply ID3 algorithm to the following Bank loan dataset to identify which CO3 (07)
feature is selected as the root node.

CASE	AGE	Has_job	Own_house	Credit rating	Class
1	Young	False	False	Good	No
2	Young	True	False	Good	Yes
3	Young	True	True	Fair	Yes
4	Young	False	False	Fair	No
5	Middle	False	False	Good	No
6	Middle	True	True	Good	Yes
7	Middle	False	True	Excellent	Yes
8	Low	False	True	Excellent	Yes
9	Low	False	True	Good	Yes
10	Low	True	False	Good	Yes

- c) List the advantages of SVM and how optimal Hyperplane differ CO3 (06)
from Hyperplane?

6. a) Write and explain the Classification Tree Algorithm. Describe the concept of 'Gini Index' as an attribute selection method. CO3 (07)
- b) Apply Naive Bayes Classifier to the below dataset and determine your activity, if you have deadlines looming, but none of them are particularly urgent, that there is no party on, and that you are currently lazy. CO3 (08)

Deadline?	Is there a party?	Lazy?	Activity
Urgent	Yes	Yes	Party
Urgent	No	Yes	Study
Near	Yes	Yes	Party
None	Yes	No	Party
None	No	Yes	Pub
None	Yes	No	Party
Near	No	No	Study
Near	No	Yes	TV
Near	Yes	Yes	Party
Urgent	No	No	Study

- c) Write K-Nearest Neighbour Algorithm? Discuss the pros and cons of the algorithms with example. CO3 (05)

UNIT- IV

7. a) Write and explain the k-means clustering algorithm. What are the four distance measures used by the classic k-means algorithm? CO4 (10)
- b) Using **AGNES** cluster the data points A (1,1), B (1.5,1.5), C (5,5), D (3,4), E (4,4), F (3,3.5). Use Euclidean distance. Draw the Dendrogram **and** the Clustering Hierarchy in X-Y space. CO4 (10)
8. a) Divide the following data into two clusters using **k-means algorithm**: Use Manhattan distance. CO4 (10)

Height	185	170	168	179	182	188	180	180	183	180	180	177
Weight	72	56	60	68	72	77	71	70	84	88	67	76

- b) Given the dataset and distance matrix in table, apply single link and complete link clustering algorithm to form clusters. CO4 (10)

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

UNIT - V

9. a) Illustrate with an example K-Fold Cross-Validation and 5×2 Cross-Validation. CO5 (10)
- b) Discuss the controllable and uncontrollable factors which affect the machine learning experiments. CO5 (10)
10. a) With an example, explain the guidelines for Machine Learning Experiments. CO5 (10)
- b) List and explain the three basic principles of machine learning experimental design. CO5 (10)
